Bento Natura

Assistant Professor · Columbia University

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304 Mudd, 500 W 120th St, New York, NY 10027

Research Interests

Algorithms, optimization, and game theory. Specific interests include linear programming, interior point methods, convex optimization, combinatorial optimization, and circuit geometry of polytopes.

Positions

Assistant Professor in Industrial Engineering & Operations Research (IEOR) *since 2024*
Columbia University, *New York City, NY*

Beerman Postdoctoral Fellow *2022 – 2024*
Georgia Institute of Technology, *Atlanta, GA*
Joint affiliations with ISyE and the College of Computing.

Postdoctoral Research Fellow *Fall 2023*
University of California, Berkeley, *Berkeley, CA*
Long Term Participant, program on *Data Structures and Optimization for Fast Algorithms* at the Simons Institute.

Postdoctoral Fellow *Spring 2023*
Brown University (ICERM), *Providence, RI*
Program: *Discrete Optimization: Mathematics, Algorithms, and Computation*.

Industry & Other

Visiting Associate, The Boston Consulting Group, *Munich, Germany* *2018*
Junior Consultant, TNG Technology Consulting, *Munich, Germany* *2018*
Research Assistant, Research Institute for Discrete Mathematics, *Bonn, Germany* *2014 – 2017*

Education

PhD in Mathematics *2018 – 2022*
London School of Economics and Political Science, *London, UK*
Supervisor: László Végh. Thesis: *Exact Linear Programming: Circuits, Curvature, and Diameter*.

MSc in Mathematics *2012 – 2017*
University of Bonn, *Bonn, Germany*
Thesis: *Algorithms for Routing and Buffer Insertion*. GPA: 4.0/4.0.

Honors & Awards

Richard Rado Prize *Oct 2024*
German Mathematical Society, *TU Berlin*
Biennial prize for the best dissertation in Discrete Mathematics.

Runner-Up for The Doctoral Award *Oct 2023*
The Operational Research Society (UK)

PhD Prize *Jul 2023*
Department of Mathematics, London School of Economics

Dodu Prize ◇ Best Talk of a Young Researcher *Sep 2020*
SMAI-MODE, *Université Paris-Saclay*

Publications

In theoretical computer science, papers are typically first published at peer-reviewed conferences. Authors are ordered alphabetically.

Journal Articles

[J7] X. Allamigeon, D. Dadush, G. Loho, **B. Natura**, L.A. Végh. "Interior Point Methods Are Not Worse Than Simplex." *SIAM Journal on Computing*, 2025. [arXiv:2206.08810]

[J6] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. "On Circuit Diameter Bounds via Circuit Imbalances." *Mathematical Programming*, 2024. [arXiv:2111.07913]

[J5] D. Dadush, S. Huiberts, **B. Natura**, L.A. Végh. "A Scaling-Invariant Algorithm for Linear Programming Whose Running Time Depends Only on the Constraint Matrix." *Mathematical Programming*, 2023. [arXiv:1912.06252]

[J4] S. Jiang, **B. Natura**, O. Weinstein. "A Faster Interior-Point Method for Sum-of-Squares Optimization." *Algorithmica*, 2023. [arXiv:2202.08489]

[J3] S. Daboul, S. Held, **B. Natura**, D. Rotter. "Global Interconnect Optimization." *ACM Trans. Design Automation of Electronic Systems*, 2023.

[J2] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. "An Accelerated Newton-Dinkelbach Method and its Application to Two Variables Per Inequality Systems." *Mathematics of Operations Research*, 2023. [arXiv:2004.08634]

[J1] F. Ekbatani, **B. Natura**, L.A. Végh. "Circuit Imbalance Measures and Linear Programming." *Surveys in Combinatorics*, 2022. [arXiv:2108.03616]

Conference Papers

[C11] **B. Natura**. "Circuit Diameter of Polyhedra is Strongly Polynomial." *FOCS*, 2026. To appear. [arXiv:2602.06958]

[C10] D. Dadush, H. Ma, **B. Natura**, L.A. Végh. "Trust Region Interior Point Methods: Optimal ℓ_2- and Faster Wide-Neighborhood Path Following." *STOC*, 2026.

[C9] D. Dadush, Z.K. Koh, **B. Natura**, N. Olver, L.A. Végh. "A Strongly Polynomial Algorithm for Linear Programs with at Most Two Non-Zero Entries per Row or Column." *STOC*, 2024. [pdf]

[C8] X. Allamigeon, D. Dadush, G. Loho, **B. Natura**, L.A. Végh. "Interior Point Methods Are Not Worse Than Simplex." *FOCS*, 2022. [arXiv:2206.08810]

[C7] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. "On Circuit Diameter Bounds via Circuit Imbalances." *IPCO*, 2022. [arXiv:2111.07913]

[C6] S. Jiang, **B. Natura**, O. Weinstein. "A Faster Interior-Point Method for Sum-of-Squares Optimization." *ICALP*, 2022. [arXiv:2202.08489]

[C5] **B. Natura**, M. Neuwohner, S. Weltge. "The Pareto Cover Problem." *ESA*, 2022. [arXiv:2202.08035]

[C4] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. "An Accelerated Newton-Dinkelbach Method and its Application to Two Variables Per Inequality Systems." *ESA*, 2021. [arXiv:2004.08634]

[C3] D. Dadush, S. Huiberts, **B. Natura**, L.A. Végh. "A Scaling-Invariant Algorithm for Linear Programming Whose Running Time Depends Only on the Constraint Matrix." *STOC*, 2020. [arXiv:1912.06252]

[C2] D. Dadush, **B. Natura**, L.A. Végh. "Revisiting Tardos's Framework for Linear Programming: Faster Exact Solutions using Approximate Solvers." *FOCS*, 2020. [arXiv:2009.04942]

[C1] S. Daboul, S. Held, **B. Natura**, D. Rotter. "Global Interconnect Optimization." *ICCAD*, 2019. [doi]

Preprints

[P2] D. Dadush, Z.K. Koh, **B. Natura**, N. Olver, L.A. Végh. "Solving Classes of Linear Programs with Strongly Polynomial Straight Line Complexity in Strongly Polynomial Time." Preprint, 2026. Full version of the initialization and bit-complexity part of [C9]. [arXiv:2610.26073]

[P1] A.N. Elmachtoub, **B. Natura**, M. Neuwohner. "Settling the Approximability of Submodular Facility Location." Preprint, 2026. [arXiv:2610.18828]

Invited Talks & Presentations

Selected Invited Lecture Series

14th Cargese Workshop on Combinatorial Optimization *Sep 2025*
INRIA/CNRS, *Corsica, France*
Lecture series on *The Combinatorics of Interior Point Methods for Linear Programming*.

Summer School on Linear Program Solvers *Jun 2025*
Georgia Institute of Technology, *Atlanta, GA*
Lecture series on *Strongly Polynomial IPM and Generalized Flows*.

"A Unified Analytical Approach to Strongly Polynomial Algorithms for Structured Linear Programs"

Oberseminar Discrete Optimization, University of Bonn, Germany *Jul 2025*

Conference & Seminar Presentations

"Circuit Diameter of Polyhedra is Strongly Polynomial"

Guest Talk, ETH Institute for Theoretical Studies, Zürich, Switzerland *Dec 2026*

Invited Talk, UC Davis, CA *Dec 2026*

INFORMS 2026, San Francisco, CA *Nov 2026*

Follow-Up Workshop: Discrete Optimization (Plenary), HIM, University of Bonn, Germany *Apr 2026*

"A strongly polynomial algorithm for the minimum-cost generalized flow problem"

Foundations of Data Science Seminar, Yale University, CT *Sep 2024*

Tutte Distinguished Lecture Series, University of Waterloo, Canada *Sep 2024*

MOPTA 2024, Lehigh University, PA *Aug 2024*

ICERM, Brown University, Providence, RI *Aug 2024*

Mixed Integer Programming Workshop (MIP 2024), Lexington, KY *Jun 2024*

Workshop on Combinatorial Optimization, Aussois, France *Jan 2024*

Special Seminar, IEOR, UC Berkeley *Nov 2023*

INFORMS 2023, Phoenix, AZ *Oct 2023*

Semester Program on Fast Algorithms, Simons Institute, UC Berkeley *Sep 2023*

"Breaking the quadratic gap for strongly polynomial solvers to combinatorial linear programs"

INFORMS 2024, Seattle, WA *Oct 2024*

ISMP 2024, Montreal, Canada *Jul 2024*

ICIAM 2023, Tokyo, Japan *Aug 2023*

CS Theory Seminar, Columbia University *Apr 2023*

CGO Seminar, London School of Economics *Dec 2022*

ACO Seminar, Georgia Institute of Technology *Nov 2022*

"Recovering strong polynomiality via straight-line-complexity"

Semester Program on Discrete Optimization, ICERM, Brown University *May 2023*

"Linear Interior Point Methods"

INFORMS Invited Optimization Society Talk, Atlanta, GA — *Mar 2026*

Korte Workshop, University of Bonn, Germany — *Nov 2025*

"Trust Region Interior Point Methods"

Discrete Optimization and Machine Learning, RIMS, Kyoto, Japan — *May 2025*

"Interior point methods are not worse than Simplex"

FOCS 2022, Denver, CO — *Nov 2022*

Young Researchers Workshop, Cornell University — *Oct 2022*

Workshop on Combinatorial Optimization, Cargese, France — *Sep 2022*

"A Faster Interior-Point Method for Sum-of-Squares Optimization"

ICALP 2022, Paris, France — *Jul 2022*

"Circuit imbalance measures and linear programming"

Focused Workshop on Integer Programming, Rényi Center, Budapest — *Aug 2023*

Workshop on Circuits and Optimization, University of Colorado Denver — *May 2023*

Semester Program Seminar, ICERM, Brown University — *Feb 2023*

Theory Seminar, University of Washington — *Mar 2022*

Northwestern Junior Theorist Workshop, Northwestern University — *Dec 2021*

Hausdorff Trimester on Discrete Optimization, University of Bonn — *Oct 2021*

"A scaling-invariant algorithm for linear programming whose running time only depends on the constraint matrix"

SMAI MODE, Université Paris-Saclay — *Sep 2020*

Research Seminar, Columbia University — *Sep 2020*

STOC 2020, Chicago, IL (virtual) — *Jun 2020*

Seminar on Combinatorics, Games and Optimisation, London School of Economics — *Dec 2019*

"Revisiting Tardos's Framework for Linear Programming"

HALG 2021, London School of Economics — *Jun 2021*

FOCS 2020, Duke University (virtual) — *Nov 2020*

Postgraduate Combinatorial Conference, Oxford University — *Jun 2019*

Teaching

Columbia University ◇ Course Instructor

Advanced Optimization (IEOR 3609) — *Spring 2025, 2026, 2027*
Convex Optimization (EEOR 6616) — *Spring 2026, 2027*
Fast Algorithms for Linear Programs (IEOR 8100) — *Spring 2025*

London School of Economics ◇ Teaching Assistant

Reinforcement Learning — TA & joint course development — *2022*
Foundations of Machine Learning — TA & joint course development — *2021, 2022*
Algorithms and Programming — *2020*
Programming in C++ — *2019*
Operational Research Methods — *2019*
Further Mathematical Methods — *2018*

University of Bonn ◇ Teaching Assistant

Combinatorics, Graphs and Matroids — *2016*
Calculus — *2013*

Service

Editorial Board ◇ *Discrete Optimization* — *since 2026*

Program Committees ◇ SODA 2025

Reviewing
Conferences: ESA ◇ FOCS ◇ ICALP ◇ IPCO ◇ ISAAC ◇ SODA ◇ STOC.
Journals: *Discrete Applied Mathematics* ◇ *Discrete Optimization* ◇ *Journal of the ACM* ◇ *Mathematical Programming* ◇ *Mathematics of Operations Research* ◇ *Networks* ◇ *Proceedings of the AMS* ◇ *SIAM Journal on Discrete Mathematics* ◇ *SIAM Journal on Optimization* ◇ *TheoretiCS*.
Grants: European Research Council (ERC).
Reviewer for *Mathematical Reviews* (MathSciNet).

Local Organizing Committee ◇ IPCO 2027, Columbia University

Seminar Organization — *since Spring 2025*
Co-organizer, IEOR-DRO Seminar, Columbia University.

Departmental Committees — *since AY 2024–25*
IEOR Department, Columbia University: PhD Admissions (2024–25, 2025–26) ◇ Undergraduate (2026–27).

Outreach — *AY 2025–26, 2026–27*
Lecture for Columbia sophomores choosing their major, IEOR Department (2025–26).
Guest lectures at Columbia Secondary School (2025–26, 2026–27).

Professional Memberships
INFORMS ◇ INFORMS Optimization Society ◇ MOS Mixed Integer Programming Society.